SQL MainQuest06_김다영

1️⃣2️⃣3️⃣4️⃣5️⃣6️⃣7️⃣8️⃣9️⃣🔟

고객 행동 기반 추천 시스템 구현하기

아이디어 출처와 링크

- ▼ 1️⃣Pattern - How to Get an Amazon's Choice Badge

  Amazon's Choice 배지는 단순히 별점이 높은 것에만 주어지지 않습니다. 알고리즘은 **낮은 반품률(Low return rate), 빠른 배송, 검색어 적합성**을 종합적으로 평가합니다. 고객의 탐색 피로도를 줄여 즉각적인 구매를 유도하는 핵심 지면

- ▼ 2️⃣Amazon - What are Amazon Lightning Deals?

  아마존의 '오늘의 딜'이나 '번개 특가(Lightning Deals)'는 제한된 시간과 재고를 시각적으로 보여주어 고객의 **FOMO(소외 두려움)**를 강하게 자극합니다. 특히 절대적인 할인 폭이 큰 고관여 상품이 이 구좌에 노출될 때 단기간에 객단가(AOV)와 판매 속도(Sales Velocity)가 폭발적으로 상승

- ▼ 3️⃣https://printify.com/blog/amazon-best-seller-badge/

  아마존의 베스트셀러 배지(주황색 리본)는 누적 데이터가 아닌 최근 판매 속도를 기준으로 매시간 업데이트됩니다. 셀러들 사이에서는 '전자기기' 같은 거대한 대분류에서 경쟁하기보다, '1m 나일론 C타입 케이블' 같은 **하위 세부 카테고리(Sub-category)를 공략**하여 1등 배지를 달고 전환율을 높이는 것이 정석적인 전략

- ▼ 4️⃣https://sell.amazon.com/programs/vine

  플랫폼의 신뢰도를 갉아먹는 가짜 리뷰를 방지하기 위해, 아마존이 직접 신뢰할 수 있는 리뷰어(Vine Voices)를 선정하여 심층 리뷰를 남기게 하는 공식 프로그램입니다. 이들이 남긴 꼼꼼한 리뷰에는 'Vine Voice'라는 특별한 초록색 텍스트가 붙어 해당 상품의 **신뢰도(Trust)**를 급상승

- ▼ 5️⃣https://www.aihello.com/resources/blog/amazon-best-seller-badge-everything-you-need-to-know/

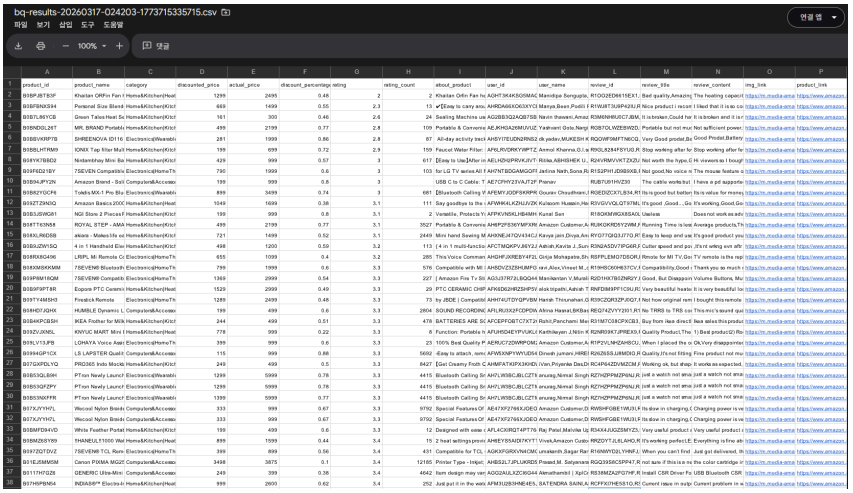
  출시된 지 90일 이내의 신상품 중 판매 추이가 좋은 상품에 'New Release' 배지를 부여합니다. 이미 리뷰가 수만 개인 기존 베스트셀러 상품들 사이에서, **초기 품질(평점)이 좋은 신규 상품에게 알고리즘 버프(노출 기회)**를 주어 새로운 셀러들이 아마존에 안착할 수 있도록 생태계를 조율

- **아마존 고급 검색 가이드 (가격 범위 필터)**https://marketaspex.com/how-to-use-amazon-advanced-search-a-complete-guide/

🌟MQ06 추천 시스템 전략 요약

#	추천 시스템 이름	가설	핵심 SQL 로직	측정 지표	아마존 실사례
①	카테고리 평균 대비 가성비 최고 상품 추천	같은 카테고리 내 평균 가격보다 저렴하면서 평점이 높은 상품은 고객에게 실질적인 가성비를 제공해 구매 전환을 유도한다	CTE로 카테고리 평균가 계산 → `discounted_price < avg_price AND rating >= 4.3` → ORDER BY price_gap DESC	price_gap (카테고리 평균가 대비 절감액)	Amazon's Choice 배지
②	오늘만 이 가격! 최대 금액 할인 VIP 추천	할인율(%)이 아닌 절대 할인 금액이 큰 상품이 고객 체감 혜택을 높여 고관여 구매를 유도한다	`(actual_price - discounted_price)` 파생 지표 생성 → ⚠️ REPLACE + CAST 전처리 필요	absolute_discount_amount (실제 절약 금액)	Today's Deals / Lightning Deals
③	세부 카테고리별 TOP 3 제패	최하위 세부 카테고리 1등 상품을 보여주면 탐색 목적이 명확한 고객의 구매 전환율이 높아진다	`UNNEST(SPLIT(category, '	rating_rank + rating_count (카테고리 내 평점 순위)	카테고리 내 전환율
④	진짜 리뷰어가 증명한 핫템	실제 텍스트 리뷰를 남긴 유저 수가 많을수록 어뷰징이 아닌 진짜 만족도를 반영해 신뢰도가 높다	`ARRAY_LENGTH(SPLIT(user_id, ','))` AS 진짜리뷰어수 → rating 결합 정렬	trust_score (rating × real_reviewer_count)	Amazon Vine / Verified Purchase
⑤	지금 비교 안 하면 손해! 가격 편차 큰 카테고리 추천	카테고리 내 가격 표준편차가 클수록 소비자의 비교 구매 필요성이 높아 추천 시스템의 가치가 극대화된다	`STDDEV()` + `HAVING COUNT() >= 5` + `ORDER BY price_stddev DESC`	price_stddev + price_range (카테고리 내 가격 분산)	Amazon Compare 기능 / 필터링 & 정렬 기능

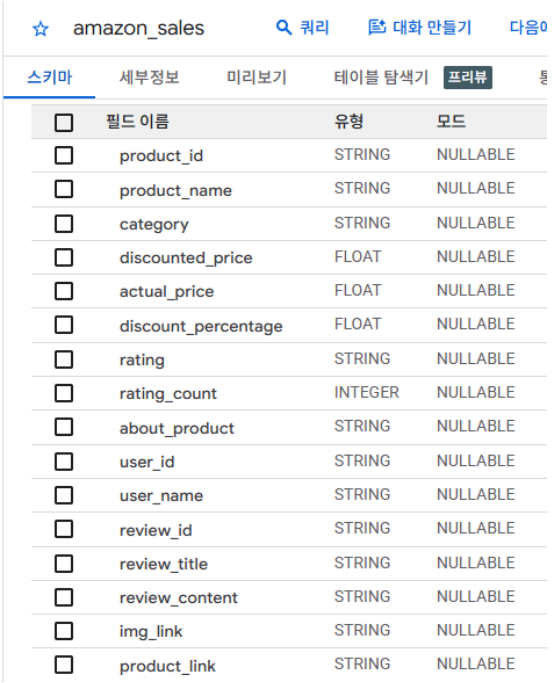
📖 **기존 csv 파일이 어떤지 확인한 스크린샷입니다.**



bq-results-20260317-024203-1773715335715.csv

	product_id	product_name	category	discounted_price	actual_price	discount_percentage	rating	rating_count	about_product	user_id	user_name	review_id	review_title	review_content	img_link	product_link
2	B08PJBT83F	Khaitan OrFin Fan h	Home&Kitchen|Heat	1299	2495	0.48	2	2	Khaitan Orfin Fan ho	AGHT3K4X50SM4C	Manodipa Sengupta,	R1OQ2ED6611EX1,	Bad quality,Amazing	The heating capacit	https://m.media-ama	https://www.amazon.
3	B08FBNKS94	Personal Size Blend	Home&Kitchen|Kitcl	669	1499	0.55	2.3	13	✔[Easy to carry ana	AHRDA6KIOE63XYC3	Manya,Been,Prabl	R1YIU8T3U6P42RJ,	Nice product	I recei	I liked that it is so ca	https://m.media-ama	https://www.amazon.
4	B07L8LY1CB	Green Tales Heat Sc	Home&Kitchen|Kitcl	161	300	0.46	2.4	24	Sealing Machine us	AGGBSQ2AQBTSS	Navin thawani,Amaz	R3MWN6U5C7J6N,	It is broken	Could her	It is broken and it is r	https://m.media-ama	https://www.amazon.
5	B08NDGL24T	MR. BRAND Portable	Home&Kitchen|Kitcl	499	2199	0.77	2.8	109	Portable & Convenie	AEJKHDA26MUVJZ	Yadhwani Gala,Nangi	RGB7OLWZEBWZD,	Portable but not muc	Not sufficient power,	https://m.media-ama	https://www.amazon.
6	B08BVKRP7B	SHREENOVA ID116	Electronics|Wearabl	281	1999	0.86	2.8	87	All-day activity track	AHSIYTEUDN2RN82	dk yadav,MUKESH K	RQOWF9MFTN6CQ,	Very Good product,Ba	Good Product,Battery	https://m.media-ama	https://www.amazon.
7	B08BLHTRM9	IONIX Tap filter Multi	Home&Kitchen|Kitcl	199	699	0.72	2.9	159	Faucet Water Filter:	AF6LRVDRKYWPTZ	Anmol Khanna,G.L.a	R9OLR24AF5YU0,R	Stop working after fe	Stop working after fe	https://m.media-ama	https://www.amazon.
8	B18YK7BB02	Nobertshop Mini Ba	Home&Kitchen|Kitcl	429	999	0.57	3	617	[Easy to Use]After in	AELAEHGPRKVJUV7	Richa,ABHISHEK U,	RQWRWVXKTZKDU	Not worth the hype,G	Hi viewers as I bought	https://m.media-ama	https://www.amazon.
9	B09F6D21BY	7SEVEN Compatible	Electronics|HomeTh	790	1999	0.6	3	103	for LG TV series All f	AH7NTBDGAMGOFT	Jatina Nath,Sona,Ri	R1S2PIHTJD98XBJ,	Not good,No voice n	The mouse feature o	https://m.media-ama	https://www.amazon.
10	B0B94JPY2N	Amazon Brand - Soli	Computers&Access	199	999	0.8	3		USB C to C Cable: T	AE7CFHY23VAJT2F	Pranav	RUBTU01HVZ30	The cable works but	I have a pd supporte	https://m.media-ama	https://www.amazon.
11	B0B2YQCP6	Toklis MX-1 Pro Blu	Electronics|Wearabl	899	3499	0.74	3	661	Bluetooth Calling V	AFEMYJODF3KRPR	Gourav Chaudhary,	R0EDIZCX7L834,R	It is good but better	It is value for money	https://m.media-ama	https://www.amazon.
12	B19ZT2RNGQ	Amazon Basics,DXX	Home&Kitchen|Heat	1549	1699	0.38	3.1	111	Say goodbye to the	AFWWK4LKZNJVZK	Kalaivani Hussain,Ha	R3VGVGJQTSTM6,	It's good ,Good....	Gio It's working,Good,Go	https://m.media-ama	https://www.amazon.
13	B0B3J5WG81	NGI Store 2 Pieces F	Home&Kitchen|Kitcl	199	999	0.8	3.1	2	Versatile, Products Y	AFPKVN5KLHBANK	Kunal Sen	R16OKWRGX8SAN,	Uselessness	Does not work as adv	https://m.media-ama	https://www.amazon.
14	B8TTK5N58	ROYAL STEP - AMA	Home&Kitchen|Kitcl	499	2199	0.77	3.1	3527	Portable & Convenie	AHEP2F5365YMXRI	Amazon Customer,A	RU9OKRD5Y2VIM,F	Running Time is low	Average products,Th	https://m.media-ama	https://www.amazon.
15	B18XLRKDSB	akara - Makes life ea	Home&Kitchen|Kitcl	721	1499	0.52	3.1	2449	Mini hand Sewing M	AH0NEJ47QV434CJ	Kavya jain,Divya,Ani	RYD77GIQ3J77O,R	Easy to keep and use	It's good product you	https://m.media-ama	https://www.amazon.
16	B0B3JZW15Q	4 in 1 Handheld Elec	Home&Kitchen|Kitcl	498	1200	0.59	3.2	113	[4 in 1 multi-functio	AFCTMQKPVJ6Y3Z	Ashish,Kavita J.,Sure	R2N2A5DV7PGFRJ	Cutter speed and poi	It's nt wking evn afte	https://m.media-ama	https://www.amazon.
17	B18R56QA56	LRIPL Mi Remote Co	Electronics|HomeTh	655	1099	0.4	3.2	285	This Voice Commen	AHGHFJ0REBY4FZ	Girija Mohapatra,Sh	R3TPLZMO7DSOR,	Rmote for Mi TV,Go	TV remote is the rep	https://m.media-ama	https://www.amazon.
18	B8XMS6KXMM	7SEVEN® Bluetooth	Electronics|HomeTh	799	1999	0.6	3.3	576	Compatible with Mi |	AHSDVZ3ZSHUMFQ	ravi,Alex,Vineet M.,r	R1IHBC6OH637CV,I	Compatibility,Good	Thank you so much	https://m.media-ama	https://www.amazon.
19	B09P8M16QM	7SEVEN® Compatible	Electronics|HomeTh	1369	2999	0.54	3.3	227	[Amazon Fire Tv Sti	AGXJ37R2LBGQ44	Manikantan V,Murali	R2D1HK7B1ZNRGY,	Good, But Disappoin	Volume Buttons, Mu	https://m.media-ama	https://www.amazon.
20	B0B9F3PT8R	Eopora PTC Ceramic	Home&Kitchen|Heat	1529	2999	0.49	3.3	29	PTC CERAMIC CHIP	AFK6DK2H6ZSHPSV	alok tripathi,Ashish T	RNFDMHPF1C9UJ,R	Very beautiful heate	It is very beautiful ho	https://m.media-ama	https://www.amazon.
21	B097Y4M6HG	Firestick Remote	Electronics|HomeTh	1289	2499	0.48	3.3	73	by JBDE | Compatibl	AHHT4UTZVQPVBM	Harish Thiruvahan,G	R3WCZQR12PJGG7,	Not how original rem	I bought this remote	https://m.media-ama	https://www.amazon.
22	B18HD7JQHX	HUMBLE Dynamic L	Computers&Access	199	499	0.6	3.3	2804	SOUND RECORDING	AFILRJ3X2FCDPDN	Afrina Hawar,BKBai	REQ7AZVYY2I01,R1	No TRRS to TRS cab	This mic's sound qua	https://m.media-ama	https://www.amazon.
23	B0B4KPCBSH	IKEA Feather for Milk	Home&Kitchen|Kitcl	244	499	0.51	3.3	478	BATTERIES ARE SC	AFCEPFGBTC7XTJI	Rohit,Panchami Mei	R31M7C08CPXCB3,	Buy from ikea direct!	Ikea sales this product	https://m.media-ama	https://www.amazon.
24	B19ZVJXN6L	KNYUC MART Mini I	Home&Kitchen|Heat	778	999	0.22	3.3	8	Function: Portable h	AFUKRD4EYPVUKLI	Karthikeyan J,Nitin K	R2NR93K7JPREX0,	I Quality Product,The	1) Best product2) Rlo	https://m.media-ama	https://www.amazon.
25	B19LV13JP9	LONAYA Voice Aide	Electronics|HomeTh	399	999	0.6	3.3	23	100% Best Quality P	AEWJCZ2WRPD4G	Amazon Customer,A	R1PZ5LMKDHR2L,	When I placed the o	Ok,Very disappointed	https://m.media-ama	https://www.amazon.
26	B19H4GP1CX	LS LAPSTER Qualit	Computers&Access	115	999	0.88	3.3	5692	-Easy to attach, remo	AFWSXNPYWYUD54	Dinesh jumani,HRE	R26ZK6SJJ8MDIO,R	Quality,It's not fitting	Fine product not mu	https://m.media-ama	https://www.amazon.
27	B07GXPDLYQ	PRO365 Indo Mouse	Home&Kitchen|Kitcl	249	499	0.5	3.3	8427	[Get Creamy Froth:	AHMFATKIPX3HDH	Ivan,Priyanka Das,Dr	R24P64ZDVMZCM,F	Working ok, but shop	It works as expected,	https://m.media-ama	https://www.amazon.
28	B0B53QL89H	PTron Newly Launch	Electronics|Wearabl	1299	5999	0.78	3.3	4415	Bluetooth Calling Sn	AHZLW3BCJBLCZT	anurag,Nimal Singh	RZ7HZPPM2P6NJ,R	just a watch not ama	just a watch not ama	https://m.media-ama	https://www.amazon.
29	B0B53QFZPY	PTron Newly Launch	Electronics|Wearabl	1299	5999	0.78	3.3	4415	Bluetooth Calling Sn	AHZLW3BCJBLCZT	anurag,Nimal Singh	RZ7HZPPM2P6NJ,R	just a watch not ama	just a watch not ama	https://m.media-ama	https://www.amazon.
30	B0B5XM7FR	PTron Newly Launch	Electronics|Wearabl	1399	5999	0.77	3.3	4415	Bluetooth Calling Sn	AHZLW3BCJBLCZT	anurag,Nimal Singh	RZ7HZPPM2P6NJ,R	just a watch not ama	just a watch not ama	https://m.media-ama	https://www.amazon.
31	B07XJYYH7L	Wecool Nylon Braide	Computers&Access	333	999	0.67	3.3	9792	Special Features:Of	AE47XF27663XJOEO	Amazon Customer,D	RW9HFGBE1WUJL,F	Its slow in charging,(	Charging power is ve	https://m.media-ama	https://www.amazon.
32	B07XJYYH7L	Wecool Nylon Braide	Computers&Access	333	999	0.67	3.3	9792	Special Features:Of	AE47XF27663XJOEO	Amazon Customer,D	RW9HFGBE1WUJL,F	Its slow in charging,(	Charging power is ve	https://m.media-ama	https://www.amazon.
33	B08MFD94VD	White Feather Portat	Home&Kitchen|Kitcl	199	499	0.6	3.3	12	Designed with ease:	AFL4CXRQT4PTT6	Raj Patel,Malvika Uj	R34XAJUQZSMYZ5,	Very useful product i	Very useful product i	https://m.media-ama	https://www.amazon.
34	B0BNQ5YT69	THANEELTL1550 Wet	Home&Kitchen|Heat	899	1599	0.44	3.4	15	2 heat settings:provi	AHBEYB4ADFKYT7	Vivek,Amazon Custo	RRZOY7JL6LAHO,R	Its working perfect,E	Everything is fine eth	https://m.media-ama	https://www.amazon.
35	B097ZQTDVZ	7SEVEN® TCL Rcm	Electronics|HomeTh	399	899	0.56	3.4	431	Compatible for TCL:	AGKX9FGRXVNHCM	umalanth,Sagar Rav	R1N6NWYD2LYHKFJ,	When you can't find	Just got delivered, th	https://m.media-ama	https://www.amazon.
36	B01EJSMMSM	Canon PIXMA MG2C	Computers&Access	3498	3875	0.1	3.4	12185	Printer Type - Inkjet;	AHBS2L7JPLUKRD5	Prasad,M. Satyanara	RGQ39S6C5PP47,R	not sure if this is re	the color cartridge ir	https://m.media-ama	https://www.amazon.
37	B1117H7GZ8	GENERIC Ultra-Mini	Computers&Access	249	399	0.38	3.4	4642	Item design may vac	AGG2AULXZCWG44	Alexatherine| | Xpl	R838MZAIFG7HF,R	Install CSR Driver Fo	USB Bluetooth CSR	https://m.media-ama	https://www.amazon.
38	B07H5PBN64	INDIAN® Electro-H	Home&Kitchen|Heat	999	2600	0.62	3.4	252	Just put it in the wall	AFM33JEB3HNE4ES,	BATENDRA SAINI,A	RCFFKIITHES51O,R3	Current issue in outp	Current problem in w	https://m.media-ama	https://www.amazon.

📖 **마찬가지로 기존 csv에 어떤 스키마가 있는지 확인한 스크린샷입니다.**



amazon_sales

쿼리 대화 만들기 다음0

스키마 세부정보 미리보기 테이블 탐색기 프리뷰

필드 이름	유형	모드
product_id	STRING	NULLABLE
product_name	STRING	NULLABLE
category	STRING	NULLABLE
discounted_price	FLOAT	NULLABLE
actual_price	FLOAT	NULLABLE
discount_percentage	FLOAT	NULLABLE
rating	STRING	NULLABLE
rating_count	INTEGER	NULLABLE
about_product	STRING	NULLABLE
user_id	STRING	NULLABLE
user_name	STRING	NULLABLE
review_id	STRING	NULLABLE
review_title	STRING	NULLABLE
review_content	STRING	NULLABLE
img_link	STRING	NULLABLE
product_link	STRING	NULLABLE

---

① 카테고리 평균 대비 가성비 최고 상품 추천

```
WITH category_avg AS (
  SELECT
    TRIM(single_category) AS single_category,
    AVG(SAFE_CAST(discounted_price AS FLOAT64)) AS avg_price
  FROM `kdy96-v02.26_03_17.amazon_sales`,
  UNNEST(SPLIT(category, '|')) AS single_category
  GROUP BY single_category
)

SELECT
  a.product_id,
  a.product_name,
  TRIM(sc) AS single_category,
  SAFE_CAST(a.discounted_price AS FLOAT64) AS price,
```

```
  ROUND(ca.avg_price, 0) AS category_avg_price,
  ROUND(ca.avg_price - SAFE_CAST(a.discounted_price AS FLOAT64), 0) AS price_gap,
  SAFE_CAST(a.rating AS FLOAT64) AS rating,
  SAFE_CAST(a.rating_count AS FLOAT64) AS rating_count
FROM `kdy96-v02.26_03_17.amazon_sales` a,
UNNEST(SPLIT(a.category, '|')) AS sc
JOIN category_avg ca ON TRIM(sc) = ca.single_category
WHERE
  SAFE_CAST(a.rating AS FLOAT64) >= 4.3
  AND SAFE_CAST(a.rating_count AS FLOAT64) >= 100
  AND SAFE_CAST(a.discounted_price AS FLOAT64) < ca.avg_price
ORDER BY
  price_gap DESC,
  SAFE_CAST(a.rating AS FLOAT64) DESC
LIMIT 10;
```

➡️

💡 비즈니스 아이디어가 나온 이유

단순히 가격이 낮은 상품이 아니라,
같은 카테고리의 다른 상품들보다 저렴한 상품을 추천해야
고객이 "이건 사야해!"라고 느낀다는 아이디어에서 출발

→ 이 느낌을 만들어주기 위해
"카테고리 평균가 vs 실제 가격" 비교 시스템을 SQL로 구현

🧠 사고 과정

- "같은 카테고리 내 평균 가격"을 먼저 구하자
  → `WITH category_avg` 에서 `AVG(discounted_price)` 로 계산
- 평균보다 저렴한 것만 → `WHERE discounted_price < ca.avg_price`
- 품질 검증된 것만 → `WHERE rating >= 4.3 AND rating_count >= 100`
- 평균 대비 얼마나 저렴한지 수치화 → `price_gap` (평균가 - 실제가)으로 만들어 `ORDER BY price_gap DESC`

📊 결과

[빅쿼리 스크린샷 아래에]
→ SmartTelevisions 카테고리 평균가(24,840루피) 대비
8,499루피짜리 VW 32인치 TV가 16,341루피 차이로 1위

⚠️ 한계

카테고리에 비정상적으로 비싼 상품(Outlier)이 섞여 있으면
평균 자체가 왜곡될 수 있음
(AVG 대신 MEDIAN 중앙값 사용 고려 필요)

🖥️[빅쿼리 결과 이미지]

Product_Price_Comparison

product_id	product_name	single_category	# price	# category_avg_price	# price_gap	# rating	# rating_count
B0B16KD737	VW 80 cm (32 inches) Playwall Frameless Series HD Ready Android Smart LED TV VW3251 (Black)	SmartTelevisions	8,499	24,840	16,341	4.3	592
B0B9XN9S3W	Acer 80 cm (32 inches) N Series HD Ready TV AR32NSV53HD (Black)	Televisions	7,999	23,305	15,306	4.3	457
B0B16KD737	VW 80 cm (32 inches) Playwall Frameless Series HD Ready Android Smart LED TV VW3251 (Black)	Televisions	8,499	23,305	14,806	4.3	592
B0B1YVCJ2Y	Acer 80 cm (32 inches) I Series HD Ready Android Smart LED TV AR32AR2841HDFL (Black)	SmartTelevisions	11,499	24,840	13,341	4.3	4,703
B0B9959XF3	Acer 80 cm (32 inches) S Series HD Ready Android Smart LED TV AR32AR2841HDSB (Black)	SmartTelevisions	12,499	24,840	12,341	4.3	1,611
B0B1YVCJ2Y	Acer 80 cm (32 inches) I Series HD Ready Android Smart LED TV AR32AR2841HDFL (Black)	Televisions	11,499	23,305	11,806	4.3	4,703
B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 32LM563BPTC (Dark Iron Gray)	SmartTelevisions	13,490	24,840	11,350	4.3	11,976
B09F6S8BT6	Samsung 80 cm (32 Inches) Wondertainment Series HD Ready LED Smart TV UA32T4340BKXXL (Glossy Black)	SmartTelevisions	13,490	24,840	11,350	4.3	16,299
B08DPLCM6T	LG 80 cm (32 inches) HD Ready Smart LED TV 32LM563BPTC (Dark Iron Gray)	SmartTelevisions	13,490	24,840	11,350	4.3	11,976
B09F6S8BT6	Samsung 80 cm (32 Inches) Wondertainment Series HD Ready LED Smart TV UA32T4340BKXXL (Glossy Black)	SmartTelevisions	13,490	24,840	11,350	4.3	16,299

---

② 오늘만 이 가격! 최대 금액 할인 VIP 추천

```
SELECT
  product_id,
  product_name,
```

```
  category,
  actual_price,
  discounted_price,
  (actual_price - discounted_price) AS absolute_discount_amount,
  rating,
  rating_count
FROM `kdy96-v02.26_03_17.amazon_sales`
WHERE
  SAFE_CAST(rating AS FLOAT64) >= 4.0
  AND SAFE_CAST(rating_count AS FLOAT64) >= 100
ORDER BY
  absolute_discount_amount DESC
LIMIT 10;
```

➡️

💡 비즈니스 아이디어가 나온 이유

단순 할인율(%)로 추천하면 1,000원짜리 케이블이 상단 독점
→ 절대 할인 금액 기준으로 바꾸면 고관여 상품이 자연스럽게 상단

🏠 실사례 — 오늘의집이 이미 하고 있었다

[영상 링크 임베드]
https://www.youtube.com/watch?v=DmlbbssojVo

> "삼성, LG 상관없이 다 합쳐서 할인해주고,
> 오프라인보다 비싸면 차액을 포인트로 보상"
> → % 할인이 아니라 절대 금액 차액 보상이 핵심

우리 SQL의 `actual_price - discounted_price` 로직과 완전히 동일한 비즈니스 원리

🧠 사고 과정

- "절대 할인 금액"이라는 파생 지표를 만들자
  → `(actual_price - discounted_price) AS absolute_discount_amount`
- 품질 검증된 상품만 → `WHERE rating >= 4.0 AND rating_count >= 100`
- 절감액 큰 것부터 → `ORDER BY absolute_discount_amount DESC`

📊 결과

[빅쿼리 스크린샷 아래에]
→ Sony Bravia TV 61,910루피 절약으로 1등

⚠️ 한계

단일 상품 기준이라 패키지 묶음 할인은 반영 못 함
(영상처럼 TV+냉장고 패키지 추천은 별도 쿼리 필요)

🖥️[빅쿼리 결과 이미지]

	A	B	C	D	E	F	G	H
1	product_id	product_name	category	actual_price	discounted_price	absolute_discou	rating	rating_count
2	B09WN3SRC7	Sony Bravia 164	Electronics|Hom	139,900	77,990	61,910	4.7	5935
3	B08GJ57MKL	Coway Professic	Home&Kitchen|H	59,900	14,400	45,500	4.4	3837
4	B08VB57558	Samsung Galax	Electronics|Mobi	74,999	37,990	37,009	4.2	27790
5	B0BB3CBFBM	VU 138 cm (55 i	Electronics|Hom	65,000	29,990	35,010	4.1	211
6	B09NS5TKPN	LG 1.5 Ton 5 Sta	Home&Kitchen|H	75,990	42,990	33,000	4.3	3231
7	B0B3XXSB1K	LG 139 cm (55 i	Electronics|Hom	79,990	47,990	32,000	4.3	1376
8	B0B19VJXQZ	ECOVACS DEE	Home&Kitchen|H	59,900	27,900	32,000	4.4	5298
9	B0BC8BQ432	VU 164 cm (65 i	Electronics|Hom	85,000	54,990	30,010	4.3	3587
10	B0B9XLX8VR	VU 139 cm (55 i	Electronics|Hom	65,000	37,999	27,001	4.3	3587
11	B08FD2VSD9	TCL 108 cm (43	Electronics|Hom	51,990	24,990	27,000	4.2	2951

➡️ Sony Bravia TV가 61,910루피 절약으로 1등

---

③ 세부 카테고리별 TOP 3 제패

```
WITH unnested AS (
  SELECT
    product_id,
    product_name,
    TRIM(single_category) AS single_category,
    SAFE_CAST(rating AS FLOAT64) AS rating,
    SAFE_CAST(rating_count AS FLOAT64) AS rating_count
  FROM `kdy96-v02.26_03_17.amazon_sales`,
  UNNEST(SPLIT(category, '|')) AS single_category
)

SELECT
  single_category,
  product_name,
  rating,
  rating_count,
  rating_rank
FROM (
  SELECT
    single_category,
    product_name,
    rating,
    rating_count,
    RANK() OVER (
      PARTITION BY single_category
      ORDER BY rating DESC, rating_count DESC
    ) AS rating_rank
  FROM unnested
  WHERE rating_count >= 100
)
WHERE rating_rank <= 3
ORDER BY single_category, rating_rank;
```

➡️

💡 비즈니스 아이디어가 나온 이유

전체 순위만 뽑으면 아이폰·맥북 같은 대형 상품이 상단 독점
→ 마이너 카테고리 고객은 추천을 아예 못 받는 문제 발생
→ 세부 카테고리별로 따로 순위를 매기면
"무선 마우스 1등", "에어프라이어 1등"처럼
어떤 카테고리 고객이든 추천을 받을 수 있음

🏆 실사례 — 아마존이 이미 하고 있다

아마존의 #1 Best Seller 배지는 전체 순위가 아니라
세부 카테고리별로 따로 붙음
"전자기기 1등"이 아니라 "1m 나일론 C타입 케이블 1등"
우리 SQL의 RANK() OVER (PARTITION BY single_category) 로직과 완전히 동일한 비즈니스 원리

🧠 사고 과정

- category가 | 로 묶여있으니 먼저 쪼개자
  → WITH unnested에서 UNNEST(SPLIT(category, '|'))
- 전체 순위가 아니라 "카테고리 안에서만" 순위를 매기자
  → RANK() OVER (PARTITION BY single_category ORDER BY rating DESC)
- 순위 붙인 결과에서 3등까지만 걸러내자
  → WHERE rating_rank <= 3 (서브쿼리로 감싸야 함)
- 리뷰 없는 상품은 제외
  → WHERE rating_count >= 100

📊 결과

[빅쿼리 스크린샷 아래에]
→ AirFryers 1등 Instant Pot(4.8), Mice 1등 Logitech M331(4.6)

→ 대분류·소분류가 둘 다 나오는 건 UNNEST로 카테고리를 쪼갠 결과
(하나의 상품이 여러 카테고리에 동시에 속할 수 있음)

⚠️ 한계

누적 평점·리뷰 수 기준이라 5년 전 출시된 고인물 상품이
영원히 1등을 차지하는 문제
(최근 1개월 리뷰 수 같은 최신성 지표 추가 필요)

🖥️[빅쿼리 결과 이미지]

	A	B	C	D	E
1	single_category	product_name	rating	rating_count	rating_rank
2	3DGlasses	Irusu Play VR Pl	3.5	621	1
3	AVReceivers&Ar	Cubetek 3 in 1 L	4	897	1
4	Accessories	10k 8k 4k HDMI	4.6	3,664	1
5	Accessories	SanDisk Extrem	4.5	205,052	2
6	Accessories	SanDisk Extrem	4.5	19,624	3
7	Accessories&Pe	Redgear MP35 S	4.6	33,434	1
8	Accessories&Pe	Logitech M331 S	4.6	12,375	2
9	Accessories&Pe	Logitech G402 H	4.6	10,760	3
10	Adapters	Amazon Basics	4.4	18,872	1
11	Adapters	rts [2 Pack] Mini	4.3	4,426	2
12	Adapters	Kanget [2 Pack]	4	1,540	3
13	Adapters&Multi-	ESnipe Mart Wo	4	2,581	1
14	AirConditioners	LG 1.5 Ton 5 Sta	4.3	3,231	1
15	AirFryers	Instant Pot Air Fi	4.8	3,964	1
16	AirFryers	INALSA Air Frye	4.5	3,192	2
17	AirFryers	PHILIPS Digital	4.4	2,981	3
18	AirPurifiers	Coway Professic	4.4	3,837	1
19	AirPurifiers	Mi Air Purifier 3	4.3	4,049	2
20	AirPurifiers	Philips Air Purifie	4.3	2,026	3
21	AirPurifiers&Ioni	Reffair AX30 [M	3.8	1,118	1
22	Arts&Crafts	Faber-Castell Co	4.3	15,867	1
23	Audio&VideoAcc	Logitech C270 D	4.3	20,398	1
24	Audio&VideoAcc	Lenovo 300 FHD	4.1	10,976	2
25	Audio&VideoAcc	ZEBRONICS Ze	4.1	5,195	3
26	AutomobileAcce	Portronics CLAM	4	18,654	1
27	AutomobileAcce	WeCool C1 Car	4	3,626	2
28	AutomobileAcce	ELV Car Mount	3.9	46,399	3
29	AutomobileChar	Duracell 38W Fa	4.4	1,680	1
30	AutomobileChar	boAt Dual Port F	4.3	38,221	2
31	AutomobileChar	Portronics CarP	4.2	4,969	3
32	BackgroundSup	DIGITEK® (DLS	4.3	6,183	1
33	Bags&Sleeves	AirCase Protecti	4.4	9,940	1
34	Bags&Sleeves	AirCase Protecti	4.3	11,330	2
35	Bags&Sleeves	AirCase Protecti	4.3	9,701	3

추천 ③ 세부 카테고리별 TOP 3 결과 (52행)

카테고리	상품명	평점	리뷰수	순위
3DGlasses	Irusu Play VR Plus Virtual Reality Headset	3.5	621	1
AVReceivers	Cubetek 3in1 LCD Bluetooth Transmitter	4.0	897	1
Accessories	10k 8k 4k HDMI Cable HDMI 2.1	4.6	3,664	1
Accessories	SanDisk Extreme SD 64GB Card	4.5	205,052	2
Accessories	SanDisk Extreme microSD 128GB	4.5	19,624	3
Accessories&Peripherals	Redgear MP35 Gaming Mousepad	4.6	33,434	1
Accessories&Peripherals	Logitech M331 Silent Wireless Mouse	4.6	12,375	2
Accessories&Peripherals	Logitech G402 Hyperion Fury Mouse	4.6	10,760	3
AirConditioners	LG 1.5 Ton 5 Star AI DUAL Inverter Split AC	4.3	3,231	1
AirFryers	Instant Pot Air Fryer Vortex 2QT	4.8	3,964	1
AirFryers	INALSA Air Fryer Digital 4L Nutri Fry	4.5	3,192	2
AirFryers	PHILIPS Digital Air Fryer HD9252	4.4	2,981	3
AirPurifiers	Coway Professional Air Purifier AirMega 150	4.4	3,837	1
AirPurifiers	MI Air Purifier 3 True HEPA Filter	4.3	4,049	2
AirPurifiers	Philips Air Purifier Ac2887	4.3	2,026	3
BluetoothSpeakers	JBL Go 2 Wireless Bluetooth Speaker	4.3	63,899	1
BluetoothSpeakers	boAt Stone 650 10W Bluetooth Speaker	4.2	41,226	2
BluetoothSpeakers	boAt Stone 250 Portable Wireless Speaker	4.2	2,908	3
Calculators	Casio MJ-12D Desktop Calculator	4.5	8,610	1
Calculators	Casio FX-991ES Plus Scientific Calculator	4.5	5,760	2
Calculators	Casio FX-82MS 2nd Gen Scientific Calculator	4.4	12,179	3
Computers&Accessories	Redgear MP35 Gaming Mousepad	4.6	33,434	1
Computers&Accessories	Logitech M331 Silent Wireless Mouse	4.6	12,375	2
Computers&Accessories	Logitech G402 Hyperion Fury Mouse	4.6	10,760	3
Electronics	Spigen EZ Fit Tempered Glass iPhone 14 Pro Max	4.7	7,779	1
Electronics	Sony Bravia 65" 4K Ultra HD Google TV	4.7	5,935	2
Electronics	Spigen EZ Fit Tempered Glass iPhone 14/13 Pro	4.6	26,603	3
GamingMice	Logitech G402 Hyperion Fury Mouse	4.6	10,760	1
GamingMice	Logitech G102 USB Light Sync Gaming Mouse	4.5	10,541	2
GamingMice	Zebronics Zeb-Transformer-M Gaming Mouse	4.4	18,139	3
Headphones	boAt BassHeads 900 On-Ear Headphones	4.2	91,188	1
Headphones	boAt Bassheads 102 Wired in Ear Earphones	4.2	76,042	2
Home&Kitchen	Swiffer Instant Electric Water Heater	4.8	53,803	1
Home&Kitchen	Instant Pot Air Fryer Vortex 2QT	4.8	3,964	2
Home&Kitchen	Campfire Spring Chef Instant Water Heater	4.7	2,591	3
Keyboards	Logitech K380 Wireless Multi-Device Keyboard	4.5	16,146	1
Keyboards	Logitech K480 Wireless Multi-Device Keyboard	4.4	22,618	2
Keyboards	Dell KB216 Wired Multimedia Keyboard	4.3	28,829	3
MemoryCards	SanDisk Extreme SD 64GB Card	4.5	205,052	1
MemoryCards	SanDisk Extreme microSD 128GB	4.5	19,624	2
MemoryCards	SanDisk Ultra SDHC 32GB	4.4	69,585	3
Mice	Logitech M331 Silent Wireless Mouse	4.6	12,375	1
Mice	Logitech Pebble M350 Wireless Mouse	4.6	10,652	2
Mice	Logitech M235 Wireless Mouse	4.5	54,405	3
MixerGrinders	Sujata Dynamix DX Mixer Grinder 900W	4.6	6,550	1
MixerGrinders	Sujata Dynamix Mixer Grinder 900W	4.6	2,751	2
MixerGrinders	Sujata Supermix Mixer Grinder 900W	4.5	1,408	3
SmartTelevisions	Sony Bravia 65" 4K Ultra HD Google TV	4.7	5,935	1
SmartTelevisions	Kodak 55" 4K Ultra HD Smart LED TV	4.4	1,712	2
SmartTelevisions	MI 32" HD Ready Android Smart LED TV	4.3	27,508	3
USBCables	AmazonBasics USB 2.0 Cable A-to-B 6ft	4.5	107,687	1
USBCables	AmazonBasics USB 2.0 Cable A-to-B 6ft	4.5	107,686	2
USBCables	AmazonBasics USB 2.0 Extension Cable 3.3ft	4.5	74,977	3

1위
2위
3위

결과값이 너무 길고 보기 불편해서 평점별로 나열하면서 우측 순위 색을 1위 초록 2위 노랑 3위 주황으로 바꿔서 이미지 재첨부했습니다.

④ 진짜 리뷰어가 증명한 핫템

🖥️[빅쿼리 결과 이미지]

⑤ 지금 비교 안 하면 손해! 가격 편차 큰 카테고리 추천

🖥️[빅쿼리 결과 이미지]